5.2 Similarity Theorems (AA, SAS~, SSS~)

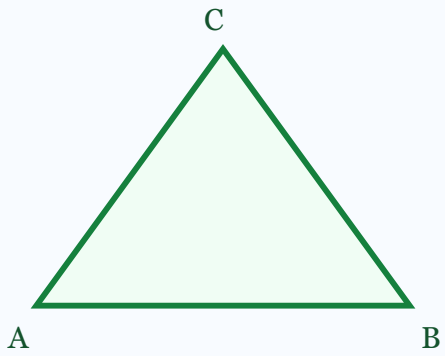
Similar triangles have the same shape but not necessarily the same size: corresponding angles are congruent, and corresponding sides are in a constant ratio (the scale factor). Three shortcuts prove similarity without checking every part. **AA (Angle-Angle):** two pairs of congruent angles are enough, since the third pair automatically matches (Angle-Sum Theorem). **SAS~ (Side-Angle-Side):** two pairs of sides in the same ratio, with the included angle congruent. **SSS~ (Side-Side-Side):** all three pairs of corresponding sides in the same ratio. A key skill is matching corresponding sides correctly — often this means sorting each triangle's sides from shortest to longest before comparing ratios. Once similarity is established, a proportion between corresponding sides (written in the order given by the similarity statement) solves for missing lengths, and perimeters of similar triangles share the same ratio as their sides.

Examples

Example 1: AA Similarity

$\angle A = 50°, \angle B = 70°$ in $\triangle ABC$; $\angle D = 50°, \angle E = 70°$ in $\triangle DEF$. Determine if the triangles are similar, and write a similarity statement.

Solution



Matching angle pairs is enough to guarantee the same shape.

Compare the given angles: $\angle A = \angle D = 50°$ and $\angle B = \angle E = 70°$ — two pairs of congruent angles.

By the Angle-Sum Theorem, the third angles must also match: $\angle C = \angle F = 180 - 50 - 70 = 60°$.

Two pairs of congruent angles is enough for similarity.

Answer: $\triangle ABC \sim \triangle DEF$ by AA.

Example 2: SSS~ — sort the sides first

$\triangle ABC$ has sides $4, 5, 7$. $\triangle DEF$ has sides $10, 14, 8$. Determine if the triangles are similar.

Solution

Sort each triangle's sides from shortest to longest so corresponding sides line up:

$\triangle ABC: 4, 5, 7.$ $\triangle DEF: 8, 10, 14.$

Form the three ratios (larger to smaller): $\frac{8}{4} = 2,\ \frac{10}{5} = 2,\ \frac{14}{7} = 2.$

All three ratios are equal.

Conclusion: the triangles are similar by SSS~, with scale factor 2.

Example 3: SAS Similarity

$MN = 4, MP = 6, \angle M = 40°$ (included) in $\triangle MNP$; $QR = 6, QS = 9, \angle Q = 40°$ (included) in $\triangle QRS$. Determine if the triangles are similar.

Solution

Check the ratio of the two given sides in each triangle:

$\frac{MN}{QR} = \frac{4}{6} = \frac{2}{3}$ and $\frac{MP}{QS} = \frac{6}{9} = \frac{2}{3}$ — equal ratios.

The included angles are congruent: $\angle M = \angle Q = 40°$.

Two pairs of proportional sides with the included angle congruent.

Conclusion: $\triangle MNP \sim \triangle QRS$ by SAS~.

Example 4: Find a missing side

$\triangle ABC \sim \triangle DEF$, with $AB = 8, DE = 12$, and $BC = 10$. Find EF.

Solution

Write a proportion using corresponding sides, in the order given by the similarity statement:

$$\frac{AB}{DE} = \frac{BC}{EF}$$

$$\frac{8}{12} = \frac{10}{EF}$$

Cross-multiply: $8 \cdot EF = 120$

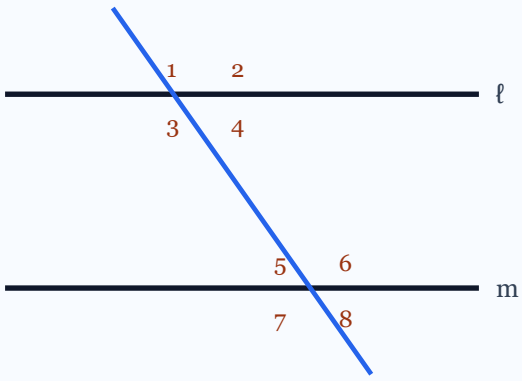
$EF = 15.$

Answer: $EF = 15.$

Example 5: Parallel line creates AA similarity

In $\triangle ABC$, $\overline{DE} \parallel \overline{BC}$ with D on $\overline{AB}$ and E on $\overline{AC}$. If $AD = 6$, $AB = 10$, $DE = 9$, find BC.

Solution



Parallel lines cut by the two transversals AB and AC create congruent corresponding angles.

Since $\overline{DE} \parallel \overline{BC}$: $\angle ADE \cong \angle ABC$ (corresponding angles, transversal $\overline{AB}$), and $\angle AED \cong \angle ACB$ (corresponding angles, transversal $\overline{AC}$).

Two pairs of congruent angles $\Rightarrow \triangle ADE \sim \triangle ABC$ by AA.

Set up the proportion: $\dfrac{AD}{AB} = \dfrac{DE}{BC}$

$\dfrac{6}{10} = \dfrac{9}{BC}$

$BC = \dfrac{9 \times 10}{6} = 15.$

Answer: $BC = 15$.

Example 6: Scale factor and perimeter

$\triangle PQR \sim \triangle XYZ$ with corresponding sides 5 and 20. If the perimeter of $\triangle PQR$ is 33, find the perimeter of $\triangle XYZ$.

Solution

Find the scale factor from $\triangle PQR$ to $\triangle XYZ$: $k = \dfrac{20}{5} = 4.$

Perimeters of similar triangles share the same ratio as their corresponding sides.

Perimeter of $\triangle XYZ = 4 \times 33 = 132.$

Answer: 132.

Example 7: Not similar (sorting reveals it)

$\triangle ABC$ has sides $3, 4, 5$. $\triangle DEF$ has sides $6, 8, 12$. Are the triangles similar?

Solution

Sort each set of sides: $3, 4, 5$ and $6, 8, 12$ (already in order).

Form the ratios: $\frac{6}{3} = 2, \frac{8}{4} = 2, \frac{12}{5} = 2.4$.

The first two ratios equal 2, but the third equals 2.4 — not all three match.

Conclusion: Not similar — SSS~ fails because the ratios are not all equal.

Example 8: Third angle confirms AA

Two triangles each have angles $40°$ and $65°$. Find the third angle in each, and state whether the triangles are similar.

Solution

Apply the Angle-Sum Theorem to each triangle: third angle $= 180 - 40 - 65 = 75°$.

Both triangles have angle sets $\{40°, 65°, 75°\}$ — all three angles match.

Conclusion: the triangles are similar by AA (only two pairs needed to be checked; the third followed automatically).

Example 9: Nested triangle — solve for x

In $\triangle ABC$, $\overline{DE} \parallel \overline{BC}$, with D on $\overline{AB}$ and E on $\overline{AC}$. $AD = 2x$, $DB = x + 3$, $AE = 8$, $EC = 10$, and $\triangle ADE \sim \triangle ABC$. Find x.

Solution

Since $\overline{DE} \parallel \overline{BC}$, $\triangle ADE \sim \triangle ABC$ by AA, so $\frac{AD}{AB} = \frac{AE}{AC}$.

$AB = AD + DB = 2x + (x + 3) = 3x + 3$; $\quad AC = AE + EC = 8 + 10 = 18$.

Set up the proportion: $\frac{2x}{3x+3} = \frac{8}{18} = \frac{4}{9}$

Cross-multiply: $9(2x) = 4(3x + 3)$

$18x = 12x + 12$

$6x = 12 \Rightarrow x = 2$.

Check: $AD = 4, AB = 9, \frac{4}{9} = \frac{8}{18}$ ✓.

Answer: $x = 2$.

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Your Turn — Show Your Work!

For each problem, solve it and verify your solution. Show all steps clearly.

Problem 1: AA

Two triangles each have angles $55°$ and $70°$. Are they similar? Why?

Answer: **Verification:**

Problem 2: SSS~

Sides $4, 6, 8$ and $6, 9, 12$. Similar?

Answer: **Verification:**

Problem 3: Missing side

$\triangle ABC \sim \triangle DEF, AB = 5, DE = 15, BC = 7$. Find EF.

Answer: **Verification:**

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Problem 4: Scale factor

Corresponding sides 8 and 2. Scale factor from larger to smaller?

Answer: **Verification:**

Problem 5: SAS~

Sides 3, 5 with included 60°; sides 6, 10 with included 60°. Similar?

Answer: **Verification:**

Problem 6: Proportion

$\triangle GHI \sim \triangle JKL, GH = 10, JK = 4, HI = 15$. Find KL.

Answer: **Verification:**

Problem 7: Not similar

Sides 5, 7, 9 and 10, 14, 20. Similar?

Answer: **Verification:**

Problem 8: Parallel

$\overline{DE} \parallel \overline{BC}$ in $\triangle ABC$, $AD = 4$, $AB = 10$, $DE = 6$. Find BC.

Answer: **Verification:**

Problem 9: AA statement

$\angle A \cong \angle D$ and $\angle B \cong \angle E$. Write a similarity statement.

Answer: **Verification:**

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Problem 10: Missing side

$\triangle RST \sim \triangle UVW$, $RS = 9$, $UV = 12$, $ST = 15$. Find VW.

Answer: **Verification:**

Problem 11: Scale factor

Sides 6, 8, 10 correspond to 9, 12, 15. Scale factor?

Answer: **Verification:**

Problem 12: Third angle

Angles 30°, 80° in one triangle; 30°, 70° in another. Similar?

Answer: **Verification:**

Problem 13: Challenge — nested triangle algebra

In $\triangle ABC$, $\overline{DE} \parallel \overline{BC}$, $AD = 3x$, $DB = x + 2$, $AE = 9$, $EC = 6$, and $\triangle ADE \sim \triangle ABC$. Find x.

Answer: **Verification:**

Problem 14: Challenge — scale factor and perimeter

$\triangle PQR \sim \triangle XYZ$. $PQ = 7$, $QR = 9$, $RP = 11$, and $XY = 21$. Find the scale factor, YZ, ZX, and the perimeter of $\triangle XYZ$.

Answer: **Verification:**

Problem 15: Challenge — sorting trap with algebra

A triangle with sides x, 12, 15 (where x is shortest) is similar to a triangle with sides 8, 20, 25. Find x.

Answer: **Verification:**

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✓ Answer Key

1. Yes; AA (matching angle pairs $55°$ and $70°$)
2. Yes; all ratios equal $\frac{3}{2}$
3. 21
4. $\frac{1}{4}$
5. Yes; SAS~
6. 6
7. No; $\frac{20}{9} \neq 2$ (the other two ratios equal 2)
8. 15
9. $\triangle ABC \sim \triangle DEF$
10. 20
11. $\frac{3}{2}$
12. No; third angles differ ($70°$ vs $80°$)
13. $x = 2$
14. Scale factor $= 3$; $YZ = 27$, $ZX = 33$; perimeter of $\triangle XYZ = 81$
15. $x = 4.8$

Great work — keep practising!